DRAFT

Paper Title

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ABSTRACT Key points of intro (context and gap). Key method. Key results. Key impact.

INDEX TERMS Magnetic sensors, Hall effect

v1.1.0+f773. Compiled on June 11, 2024 at 11:17.

> **Examples**
>
> Here are two of my recent papers formatted with Quarto:
>
> - [1] is a tutorial paper on technical writing with Quarto.
> - [2] is a full technical paper, whose source is available on Gitlab.
>
> Note that there is now an official extension to format the output according to IEEE template

I. INTRODUCTION

A. CONTEXT

- Big-picture challenge in the world ("hook the audience").
- Zoom into a specific topic.
- Prior knowledge & work, scientific SoA
- Open research question & gap
  - there is a **need** to address this gap

B. OBJECTIVES

Or "Sensor requirements"

- High-level restrictions (IN vs OUT of scope)
- Specific measurable metrics

C. PREVIEW

- Preview the novel idea (concept, approach, ...) TIP: include a concept overview figure on page 1. See Figure 1.
- Highlight unique contributions.
- Preview paper structure: Section 1 describes ...

Figure 1: Key concept.

II. (RELATED WORKS)

In long paper, dedicated section to review the state of the art.

III. METHODS

Or "Design" if more appropriate. Use action verb to emphasize the procedure (the "recipe").

Figure 2 shows the architecture block diagram.

A. DESIGN OF ELEMENT 1

B. DESIGN OF ELEMENT 2

IV. IMPLEMENTATION

Pictures of realizations and/or experimental setups.

Figure 2: Block diagram.

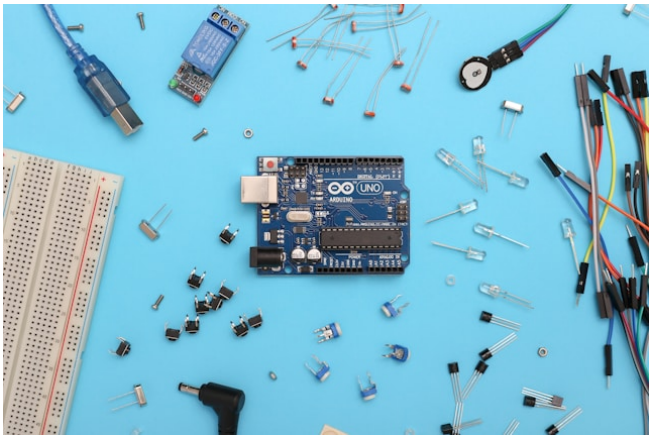

Figure 3: Picture of the prototype.

V. (EXPERIMENTAL) RESULTS

- Test 1
- Test 2

VI. DISCUSSION

- Key insights
- Comparison with previous work
- Highlight the strengths by explaining
  - how the paper moves the SoA
  - how the paper addresses the original objectives
- Discuss limitations and recommendation for future work

VII. CONCLUSION

- Summary: methods & results & significance
- Extrapolate / speculate on the broad impact

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APPENDIX A: REFRESHER ON XXX

It might be needed to provide a refresher on some well-known background knowledge. Review of Theory X, Fundamentals of X, ...

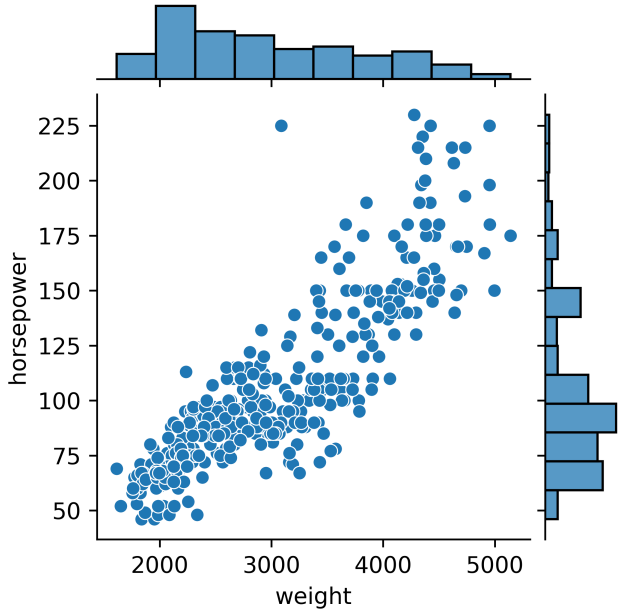

Figure 4: Result of test 1.

APPENDIX B: NOMENCLATURE / NOTATION

List of symbols. Sometimes included just after the intro.

APPENDIX C: REFERENCES

[1] G. Close, "Technical Writing and Publishing Data-Rich Articles with Quarto," Sep. 22, 2022. [Online]. Available: https://gael-close.github.io/posts/2209-tech-writing/2209-tech-writing.html

[2] B. Brajon, E. Gasparin, and G. Close, "A benchmark of integrated magnetometers and magnetic gradiometers," IEEE Access, vol. 11, pp. 115635–115643, 2023, doi: 10.1109/ACCESS.2023.3325035.